Package 'CIMEHR'

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Title Gaussian clinically informative visiting and observation processes in Electronic Health Record (EHR) data

Version 0.1.0

Description Fits semiparametric joint models for longitudinal electronic health record (EHR) data that addresses two-stage hierarchical missingness mechanism. The first stage is the visiting process, and the second stage is the observation process. The core GIVEHR method (Generalized Informative Visiting and observation processes in Electronic Health Records) uses a three-stage procedure: partial likelihood with log-normal frailty for visit intensity, probit regression with shared latent factor-linked random effects for observation, and weighted least squares with risk-set centering for the ouctome. These three stages are connected through a shared latent factor that induces dependence across all three processes. A data simulator and implementations of common benchmark methods (linear midex models, multiple imputation, and others) are included for comparative studies. Detailed methods are described in Yang, Shi, and Mukherjee (2026) <https://doi.org/10.48550/arXiv.2602.15374>.

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Encoding UTF-8

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stats,
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data.table,
mice,
nlme,
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LinkingTo Rcpp

URL https://github.com/ysph-dsde/CIMEHR

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VignetteBuilder knitr

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LazyData true

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bootstrap *Bootstrap Confidence Intervals for CIMEHR Model Fits*

Description

Computes bootstrap standard errors and percentile confidence intervals by resampling subjects with replacement and re-fitting the original model. Suitable for all CIMEHR methods that do not provide closed-form standard errors: GIVEHR_GH, EHRJoint, JMVL_Liang, IIRR_weighting, and IIRR_balancing. Can also supplement the sandwich standard errors from GIVEHR and GIVEHR_OU_PCL.

Usage

```
bootstrap(
  fit,
  data,
  id = "id",
  B = 1000,
  level = 0.95,
  covars = list(),
  which_estimate = c("bal", "mle"),
  seed = NULL,
  verbose = FALSE
)
```

Arguments

fit	A fitted model object from one of the CIMEHR method functions. Supported classes: "GIVEHR", "GIVEHR_GH", "GIVEHR_OU_PCL", "EHRJoint", "JMVL_Liang", "IIRR_weighting", "IIRR_balancing".
data	The data.frame used to fit fit, in long (panel) format with one row per subject-visit.
id	Character string giving the name of the subject identifier column. Default is "id".
B	Integer; number of bootstrap replicates. Default is 1000.
level	Confidence level for percentile intervals. Default is 0.95.
covars	Named list of model-specific arguments forwarded to the re-fitting function on each replicate. Supply only arguments that differ from the function defaults; unspecified arguments use the defaults listed in the Details section. Names must match the argument names of the underlying fitting function exactly; unrecognised names trigger a warning.
which_estimate	For "IIRR_balancing" fits only: which estimator to display in print and summary. One of "bal" (default, balancing weights) or "mle" (MLE weights). Both estimators are always stored in boot_matrix with column prefixes "bal." and "mle.".
seed	Integer; random seed for reproducibility. Default is NULL (no seed set).
verbose	Logical; if TRUE, a progress message is printed every 100 replicates. Default is FALSE.

Details

For two-stage methods (LI_IP, MI_IP), the bootstrap must cover the full two-stage procedure. Use those functions directly inside a bootstrap loop rather than this wrapper.

Subjects are resampled with replacement. For each replicate, all rows belonging to a sampled subject are included; subjects drawn multiple times receive distinct integer IDs to avoid duplicate identifiers in the resampled dataset.

Replicates that throw an error or a convergence-related warning are silently discarded. A warning is issued if more than 10\ fail.

Value

An object of class "cimehr_bootstrap", a named list with:

estimates Named numeric vector of point estimates from the original fit (outcome coefficients of the displayed estimator).

boot_matrix Numeric matrix of dimension B_success by p; each row is the outcome coefficient vector from one bootstrap replicate. For "IIRR_balancing", columns are prefixed "bal." and "mle.".

se Named numeric vector of bootstrap standard errors for the displayed estimator.

ci Numeric matrix with columns CI.low and CI.high; percentile intervals for the displayed estimator.

level The confidence level used.

B Number of replicates requested.

B_success Number of replicates that converged.

method Character string; class of the original fit.

which_estimate For "IIRR_balancing": which estimator is shown in print/summary ("bal" or "mle"). NULL for all other methods.

call The matched call.

Covariate arguments by method

Supply only entries in covars that differ from the defaults shown below. Names must match the argument names of the underlying fitting function exactly.

GIVEHR, GIVEHR_GH, GIVEHR_OU_PCL:

covars_visit_XV Default c("Z", "X").

covars_outcome_fixed_XY Default c("Z", "X").

covars_outcome_link_ZY Default "Z".

covars_obs_fixed_XO Default c("Z", "X").

covars_obs_random_ZO Default "Z".

EHRJoint, JMVL_Liang:

visit_covs Default c("Z", "X").

long_covs Default c("Z", "X").

random_covs Default "Z".

IIRR_weighting:

`visit_covs` Default `c("Z", "X")`.

`formula` Default `NULL` (auto-constructed from `visit_covs` and `time`).

IIRR_balancing:

`visit_covs` Default `c("Z", "X")`.

`balance_covs` Default `NULL` (uses `visit_covs`).

`formula` Default `NULL` (auto-constructed).

`phi` Sensitivity parameter. Default `0`.

Plug-in vs. bootstrap

`GIVEHR` and `GIVEHR_OU_PCL` already compute sandwich (plug-in) standard errors from per-subject score contributions evaluated at the fitted parameters — no re-fitting is needed. Bootstrap is provided for those methods as an optional check. For `GIVEHR_GH`, `EHRJoint`, `JMVL_Liang`, `IIRR_weighting`, and `IIRR_balancing`, per-subject score contributions are not exposed by the fitting routines, so bootstrap is the recommended approach for uncertainty quantification.

References

Bradley Efron and Robert J. Tibshirani (1994). *An Introduction to the Bootstrap*. Chapman & Hall/CRC.

See Also

`print.cimehr_bootstrap`, `summary.cimehr_bootstrap`

Examples

```
dat <- sim_ehr_data[sim_ehr_data$id <= 100, ]
fit <- IIRR_weighting(dat, visit_covs = c("Z", "X"))
bs  <- bootstrap(fit, data = dat, B = 5, seed = 1)
print(bs)
```

`coef.GIVEHR` *Extract Coefficients from a GIVEHR Fit*

Description

Extract Coefficients from a GIVEHR Fit

Usage

```
## S3 method for class 'GIVEHR'
coef(object, type = c("outcome", "visiting", "observation"), ...)

## S3 method for class 'GIVEHR_GH'
coef(object, type = c("outcome", "visiting", "observation"), ...)

## S3 method for class 'GIVEHR_OU_PCL'
coef(object, type = c("outcome", "visiting", "observation"), ...)
```

Arguments

object	A GIVEHR, GIVEHR_GH, or GIVEHR_OU_PCL object.
type	Which coefficients: "outcome" (default), "visiting", or "observation".
...	Ignored.

Value

Named numeric vector.

Examples

```
dat <- sim_ehr_data[sim_ehr_data$id <= 80, ]
fit <- GIVEHR(data = dat)
coef(fit)
coef(fit, type = "visiting")
```

---

confint.GIVEHR *Confidence Intervals for GIVEHR Outcome Coefficients*

Description

Computes Wald confidence intervals using sandwich standard errors.

Usage

```
## S3 method for class 'GIVEHR'
confint(object, parm = NULL, level = 0.95, ...)

## S3 method for class 'GIVEHR_OU_PCL'
confint(object, parm = NULL, level = 0.95, ...)
```

Arguments

object	A GIVEHR or GIVEHR_OU_PCL object.
parm	Character vector of parameter names, or NULL for all.
level	Confidence level. Default is 0.95.
...	Ignored.

Value

Numeric matrix with lower and upper bounds, or NULL if standard errors are unavailable.

Examples

```
dat <- sim_ehr_data[sim_ehr_data$id <= 80, ]
fit <- GIVEHR(data = dat)
confint(fit)
confint(fit, parm = "Z", level = 0.90)
```

`EHRJoint` *Joint Estimation for Electronic Health Record (EHR) Longitudinal Processes (EHRJoint)*

Description

Fit a joint workflow for visiting, observation, and outcome processes for irregularly observed longitudinal data.

Usage

```
EHRJoint(
  data,
  id = "id",
  time = "time",
  outcome = "Y",
  obs_indicator = "R",
  visit_covs = c("Z", "X"),
  long_covs = c("Z", "X"),
  random_covs = "Z"
)
```

Arguments

`data`	A `data.frame` containing longitudinal records.
`id`	Character string. Subject identifier column name. Default is `"id"`.
`time`	Character string. Time column name. Default is `"time"`.
`outcome`	Character string. Outcome column name. Default is `"Y"`.
`obs_indicator`	Character string. Observation indicator column name. Default is `"R"`.
`visit_covs`	Character vector of covariate names for the visiting model. Default is `c("Z", "X")`.
`long_covs`	Character vector of covariate names for the longitudinal outcome model. Default is `c("Z", "X")`.
`random_covs`	Character string or vector of covariate names that interact with the latent factor. Default is `"Z"`.

Value

An object of class `"EHRJoint"` containing component-wise coefficient estimates. If available, standard errors are stored and used by `summary()` to form 95\

References

Jiacong Du, Xu Shi, and Bhramar Mukherjee (2025). A new statistical approach for joint modeling of longitudinal outcomes measured in electronic health records with clinically informative presence and observation processes. arXiv preprint arXiv:2410.13113. doi:10.48550/arXiv.2410.13113. https://arxiv.org/abs/2410.13113.

Examples

```
dat <- sim_ehr_data[sim_ehr_data$id <= 80, ]
fit <- EHRJoint(dat)
print(fit)
```

GIVEHR *Generalized Informative Visiting and observation processes in Electronic Health Records (GIVEHR)*

Description

Fits a three-stage semiparametric joint model for longitudinal Electronic Health Record (EHR) data that simultaneously accounts for an informative visiting process and an informative observation process. Stage 1 estimates the visiting-process intensity via partial likelihood with a log-normal frailty. Stage 2 models the observation indicator with a probit regression that may include latent-factor-linked random effects. Stage 3 estimates the longitudinal outcome coefficients by Weighted Least Squares (WLS) with risk-set centering. Sandwich variance estimators are provided for all three stages.

Usage

```
GIVEHR(
  data,
  id_col = "id",
  time_col = "time",
  y_col = "Y",
  r_col = "R",
  censor_col = "C",
  covars_visit_XV = c("Z", "X"),
  covars_outcome_fixed_XY = c("Z", "X"),
  covars_outcome_link_ZY = c("Z"),
  covars_obs_fixed_XO = c("Z", "X"),
  covars_obs_random_ZO = "Z",
  time_precision = 0.01
)
```

Arguments

data	A data.frame in long (panel) format with one row per subject-time combination. Must contain columns for subject identifier, visit time, outcome, observation indicator, censoring time, and all specified covariates. Rows with NA in the time column are treated as unvisited time points.
id_col	Character string giving the name of the subject identifier column in data. Default is "id".
time_col	Character string giving the name of the visit-time column in data. Default is "time".
y_col	Character string giving the name of the outcome column in data. Default is "Y".

`r_col` Character string giving the name of the binary observation indicator column (`1` = outcome observed at visit, `0` = outcome missing at visit). Default is `"R"`.

`censor_col` Character string giving the name of the column containing each subject's censoring (end-of-follow-up) time. One value per subject is expected; the value on the first row for each subject is used. Default is `"C"`.

`covars_visit_XV`
Character vector of covariate column names for the visiting-process intensity model. These enter the exponent of the multiplicative intensity. Default is `c("Z", "X")`.

`covars_outcome_fixed_XY`
Character vector of covariate column names for the fixed-effect part of the longitudinal outcome model. No intercept is added automatically. Default is `c("Z", "X")`.

`covars_outcome_link_ZY`
Character vector of covariate column names that interact with the conditional expectation of the latent factor U_i in the outcome model. An intercept column is prepended automatically. Including `time_col` in this vector adds a time-varying latent link. Default is `c("Z")`.

`covars_obs_fixed_XO`
Character vector of covariate column names for the fixed-effect part of the observation-process probit model. An intercept is prepended automatically. Default is `c("Z", "X")`.

`covars_obs_random_ZO`
Character vector of covariate column names that receive independent random effects in the observation process. An intercept is prepended automatically. Set to `NULL` or an empty vector to fit a fixed-effects-only probit. Default is `"Z"`.

`time_precision` Positive numeric scalar for time discretisation. Visit times are rounded to the nearest multiple of `time_precision` before constructing risk sets. Set to `NULL` or `0` to skip rounding. Default is `0.01`.

Details

The GIVEHR estimator proceeds in three stages:

Stage 1 - Visiting Process. The recurrent-event intensity for subject i is modelled as

$$\lambda_i(t) = \eta_i \exp(\gamma^\top X_i^V)\, \lambda_0(t),$$

where $\eta_i = \exp(\mu_0 + \sigma_\zeta U_i)$ is a log-normal frailty with $U_i \sim N(0,1)$ and $\mu_0 = -\sigma_\zeta^2/2$ ensures $E[\eta_i] = 1$. The coefficient vector γ is estimated by Newton-Raphson on the partial likelihood score, the baseline hazard λ_0 by the Breslow estimator, and the frailty variance σ_ζ^2 by a method-of-moments estimator. Subject-level posterior summaries $(\hat{\mu}_{U_i}, \hat{\sigma}^2_{U_i})$ are obtained via Laplace approximation. A sandwich estimator provides inference for γ.

Stage 2 - Observation Process. At each visit the probability that the outcome is recorded is modelled as

$$P(R_{ij} = 1 \mid U_i, q_i) = \Phi\big(\alpha^\top X_{ij}^O + \delta^\top Z_{ij}^O \cdot U_i + q_i^\top Z_{ij}^O\big),$$

where $q_i \sim N(0, \Sigma_q)$ are independent random effects and Φ is the standard-normal CDF. The marginal log-likelihood (integrated over q_i using the empirical Bayes posterior of U_i) is maximised by `nlminb` with analytic gradients. When the estimated frailty variance is negligible or no random covariates are specified, a fixed-effects probit is fitted by iteratively reweighted least squares. A clustered sandwich estimator provides standard errors.

Stage 3 - Longitudinal Outcome. The outcome model is

$$Y_i(t) = \beta^\top X_i^Y(t) + \theta^\top Z_i^Y(t)\,\kappa_i(t) + \varepsilon_i(t),$$

where $\kappa_i(t) = E[U_i \mid \text{observation history up to } t]$ is the conditional expectation of the latent factor given the observation history. Estimation uses weighted least squares with time-varying risk-set centering, where the weights combine the marginal observation probability $\varpi_i(t)$ and the visit intensity. A sandwich estimator provides standard errors for β.

Value

A named list with three components, each itself a named list:

`visiting_process` A list with elements:

- `gamma_V` Named numeric vector of visiting-process regression coefficients from partial likelihood.
- `se_gamma` Named numeric vector of sandwich standard errors for `gamma_V`.
- `vcov_gamma` Numeric matrix; the sandwich variance-covariance matrix for `gamma_V`.
- `Lambda0_hat` Named numeric vector of Breslow baseline hazard increments $d\hat{\Lambda}_0(t_k)$ at each unique discretised event time.
- `sigma2_zeta` Numeric scalar; estimated log-normal frailty variance on the log scale.
- `mu_Ui` Named numeric vector of empirical Bayes posterior means for the standardised latent factor U_i.
- `sigma2_Ui` Named numeric vector of empirical Bayes posterior variances for U_i.
- `Ci` Named numeric vector of per-subject censoring times.
- `mi` Named integer vector of per-subject visit counts.

`observation_process` A list with elements:

- `alpha_fixed` Named numeric vector of fixed-effect probit coefficients (including intercept).
- `se_alpha` Named numeric vector of sandwich standard errors for `alpha_fixed`.
- `vcov_obs` Numeric matrix; sandwich variance-covariance matrix for all observation-process parameters.
- `delta_latent_link` (Present only when random effects are fitted.) Named numeric vector of latent-link coefficients δ.
- `se_delta` (Present only when random effects are fitted.) Named numeric vector of sandwich standard errors for `delta_latent_link`.
- `Sigma_q` (Present only when random effects are fitted.) Named numeric vector of estimated random-effect standard deviations.

`longitudinal_outcome` A list with elements:

- `beta_Y` Named numeric vector of outcome fixed-effect coefficients.
- `theta_Y` Named numeric vector of outcome latent-link coefficients.
- `se_beta_naive` Named numeric vector of sandwich standard errors for `beta_Y`.
- `vcov_joint` Numeric matrix; sandwich variance-covariance matrix for the full coefficient vector `c(beta_Y, theta_Y)`.

References

Yang, C.-H., Shi, X., and Mukherjee, B. (2026). "Joint Modeling of Longitudinal EHR Data with Shared Random Effects for Informative Visiting and Observation Processes." arXiv:2602.15374. doi:10.48550/arXiv.2602.15374.

See Also

GIVEHR_GH for a Gauss-Hermite quadrature variant with online filtering; GIVEHR_OU_PCL for the Ornstein-Uhlenbeck pairwise composite likelihood extension; sim_data_gen for simulating data compatible with this model.

Examples

```
data(sim_ehr_data)
fit <- GIVEHR(data = sim_ehr_data)
print(fit)
summary(fit)
coef(fit)
confint(fit)
```

GIVEHR_GH *Generalized Informative Visiting and Observation Processes in Electronic Health Records (GIVEHR) with Gauss–Hermite (GH) Quadrature*

Description

Fits a three-stage semiparametric joint model for longitudinal Electronic Health Record (EHR) data, extending GIVEHR by replacing the closed-form marginal observation-probability approximation with Gauss–Hermite (GH) quadrature and an online Laplace filter. This enables exact integration over both the shared latent factor U_i and the observation-process random effects q_i, and supports a time-varying observation process through a lagged observation indicator.

Stage 1 estimates the visiting-process intensity via Andersen–Gill partial likelihood with a log-normal frailty. Stage 2 maximizes a Gauss-Hermite-integrated marginal log-likelihood for the probit observation model. Stage 3 estimates the longitudinal outcome coefficients by weighted least squares with time-varying risk-set centering and sequentially updated $\omega_i(t)$ and $\kappa_i(t)$.

Usage

```
GIVEHR_GH(
  data,
  id_col = "id",
  time_col = "time",
  y_col = "Y",
  r_col = "R",
  r_lag_col = NULL,
  covars_visit_XV = c("Z", "X"),
  covars_outcome_fixed_XY = c("Z", "X"),
  covars_outcome_link_ZY = c("Z"),
  covars_obs_fixed_XO = c("Z", "X"),
  covars_obs_random_ZO = "Z",
  gh_points_U = 7,
  gh_points_q = 5,
  gh_clip_eta = 12,
  clip_k = 12,
```

```
  filter_max_iter = 6,
  verbose = FALSE
)
```

Arguments

data	A data.frame in long (panel) format with one row per subject-time combination. Must contain columns for subject identifier, visit time, outcome, observation indicator, and all specified covariates. Rows with NA in the time column are treated as unvisited time points.
id_col	Character string giving the name of the subject identifier column. Default is "id".
time_col	Character string giving the name of the visit-time column. Default is "time".
y_col	Character string giving the name of the outcome column. Default is "Y".
r_col	Character string giving the name of the binary observation indicator column (1 = outcome observed, 0 = outcome missing at visit). Default is "R".
r_lag_col	Character string giving the name of a lagged observation indicator column. When non-NULL, a time-varying coefficient λ on R_{i,j-1} is added to the observation-process linear predictor, allowing the probability of observation to depend on whether the outcome was observed at the previous visit. When NULL (the default), the lagged indicator is constructed internally with first-visit values set to zero, but the λ coefficient is omitted from the model.
covars_visit_XV	Character vector of covariate column names for the visiting-process intensity model. Default is c("Z", "X").
covars_outcome_fixed_XY	Character vector of covariate column names for the fixed-effect part of the longitudinal outcome model. No intercept is added automatically. Default is c("Z", "X").
covars_outcome_link_ZY	Character vector of covariate column names that interact with the conditional expectation of the latent factor U_i in the outcome model. An intercept is prepended automatically. Including time_col adds a time-varying link. Default is c("Z").
covars_obs_fixed_XO	Character vector of covariate column names for the fixed-effect part of the observation-process probit model. An intercept is prepended automatically. Default is c("Z", "X").
covars_obs_random_ZO	Character vector of covariate column names receiving independent random effects in the observation process. An intercept is prepended automatically. Set to NULL or an empty vector for a fixed-effects-only probit. Default is "Z".
gh_points_U	Integer giving the number of Gauss–Hermite quadrature nodes for integrating over the standardised latent factor U_i. More nodes improve accuracy at the cost of speed. Default is 7.
gh_points_q	Integer giving the number of Gauss–Hermite quadrature nodes for each dimension of the observation-process random effects q_i. A tensor-product grid is formed when q_i is multi-dimensional. Default is 5.
gh_clip_eta	Numeric scalar; linear predictor values in the probit model are clipped to [-gh_clip_eta, gh_clip_eta] for numerical stability during Stage 2 optimisation. Default is 12.

`clip_k` Numeric scalar; the standardised argument to Φ is clipped to `[-clip_k, clip_k]` when computing marginal observation probabilities $\omega_i(t)$ and conditional expectations $\kappa_i(t)$. Default is `12`.

`filter_max_iter` Integer giving the maximum number of Newton iterations for the Laplace approximation used in the online filter that updates the joint posterior of (U_i, q_i) after each observation event. Default is `6`.

`verbose` Logical; if `TRUE`, progress messages are printed during estimation. Default is `FALSE`.

Details

The model structure is the same as `GIVEHR` but the computational strategy differs in two key respects:

Stage 2 – Gauss–Hermite Quadrature. Instead of the Laplace-style marginal approximation used in `GIVEHR`, the marginal observation-process log-likelihood is evaluated by Gauss–Hermite quadrature over U_i (centred on the empirical Bayes posterior from Stage 1) and over q_i. The quadrature grid has `gh_points_U` nodes for U_i and a tensor-product grid of `gh_points_q` nodes per dimension for q_i. The marginal negative log-likelihood is minimised with `nlminb`. When `r_lag_col` is specified, an additional parameter λ captures dependence of observation on the previous observation status.

Stage 3 – Online Filtering. The marginal observation probability $\omega_i(t)$ and conditional expectation $\kappa_i(t)$ are computed via a sequential (online) Laplace filter that updates the joint posterior of (U_i, q_i) after each observation event. This produces time-varying versions of ω and κ that condition on the full observation history up to each time point, rather than treating them as static subject-level quantities.

The visiting-process estimation (Stage 1) is identical to `GIVEHR`.

Value

A named list with three components:

`visiting_process` A list with elements:

- `gamma_V` Named numeric vector of visiting-process regression coefficients (Andersen–Gill partial likelihood).
- `Lambda0_C` Numeric scalar; Breslow estimate of the cumulative baseline hazard over the full follow-up window.
- `sigma2_zeta` Numeric scalar; estimated log-normal frailty variance on the log scale.
- `mu_Ui` Named numeric vector of empirical Bayes posterior means for the standardised latent factor U_i.
- `sigma2_Ui` Named numeric vector of empirical Bayes posterior variances for U_i.

`observation_process` A list with elements:

- `alpha_fixed` Named numeric vector of fixed-effect probit coefficients (including intercept).
- `delta_latent_link` (Present only when random effects are fitted.) Named numeric vector of latent-link coefficients δ.
- `Sigma_q` (Present only when random effects are fitted.) Numeric matrix; estimated random-effect covariance matrix (diagonal).
- `lambda` (Present only when `r_lag_col` is non-`NULL`.) Numeric scalar; estimated coefficient for the lagged observation indicator.

`longitudinal_outcome` A list with elements:

- `beta_Y` Named numeric vector of outcome fixed-effect coefficients.
- `theta_Y` Named numeric vector of outcome latent-link coefficients.

Note

Computation time increases with gh_points_U, gh_points_q, and the number of subjects. For very large datasets, consider using GIVEHR (which avoids quadrature) as a faster alternative.

References

Yang, C.-H., Shi, X., and Mukherjee, B. (2026). "Joint Modeling of Longitudinal EHR Data with Shared Random Effects for Informative Visiting and Observation Processes." arXiv:2602.15374. doi:10.48550/arXiv.2602.15374.

See Also

GIVEHR for the base version without quadrature; GIVEHR_OU_PCL for the Ornstein-Uhlenbeck pairwise composite likelihood extension; sim_data_gen for simulating compatible data.

Examples

```
dat <- sim_ehr_data[sim_ehr_data$id <= 50, ]
fit <- GIVEHR_GH(data = dat, gh_points_U = 3, gh_points_q = 3)
print(fit)
```

GIVEHR_OU_PCL *Generalized Informative Visiting and Observation Processes in Electronic Health Records (GIVEHR) with Ornstein–Uhlenbeck (OU) Pairwise Composite Likelihood (PCL) Extension*

Description

Fits a three-stage semiparametric joint model for longitudinal Electronic Health Record (EHR) data, extending GIVEHR by modelling temporal serial correlation in the observation process with a stationary Ornstein–Uhlenbeck (OU) process and estimating the observation-process parameters via Pairwise Composite Likelihood (PCL).

Stage 1 estimates the visiting-process intensity via Andersen–Gill partial likelihood with a log-normal frailty (identical to GIVEHR). Stage 2 maximises a pairwise composite log-likelihood over adjacent (or all) within-subject visit pairs, integrating over the bivariate probit structure induced by the OU process. An analytic gradient is provided for efficiency. Stage 3 estimates the longitudinal outcome coefficients by weighted least squares with risk-set centering and a robust sandwich variance estimator.

Usage

```
GIVEHR_OU_PCL(
  data,
  id_col = "id",
  time_col = "time",
  y_col = "Y",
  r_col = "R",
  covars_visit_XV = c("Z", "X"),
  covars_outcome_fixed_XY = c("Z", "X"),
  covars_outcome_link_ZY = c("Z"),
```

```
  covars_obs_fixed_XO = c("Z", "X"),
  covars_obs_random_ZO = "Z",
  pair_type = c("adjacent", "all"),
  phi_init = 0.1,
  phi_bounds = c(1e-04, 100),
  optim_method = "L-BFGS-B",
  verbose = FALSE
)
```

Arguments

data	A `data.frame` in long (panel) format with one row per subject-time combination. Must contain columns for subject identifier, visit time, outcome, observation indicator, and all specified covariates. Rows with `NA` in the time column are treated as unvisited time points.
id_col	Character string giving the name of the subject identifier column. Default is `"id"`.
time_col	Character string giving the name of the visit-time column. Default is `"time"`.
y_col	Character string giving the name of the outcome column. Default is `"Y"`.
r_col	Character string giving the name of the binary observation indicator column (`1` = outcome observed, `0` = outcome missing at visit). Default is `"R"`.
covars_visit_XV	Character vector of covariate column names for the visiting-process intensity model. Default is `c("Z", "X")`.
covars_outcome_fixed_XY	Character vector of covariate column names for the fixed-effect part of the longitudinal outcome model. No intercept is added automatically. Default is `c("Z", "X")`.
covars_outcome_link_ZY	Character vector of covariate column names that interact with the conditional expectation of the latent factor U_i in the outcome model. An intercept is prepended automatically. Including `time_col` adds a time-varying link. Default is `c("Z")`.
covars_obs_fixed_XO	Character vector of covariate column names for the fixed-effect part of the observation-process probit model. An intercept is prepended automatically. Default is `c("Z", "X")`.
covars_obs_random_ZO	Character vector of covariate column names receiving independent random effects in the observation process. An intercept is prepended automatically. Set to `NULL` for a fixed-effects-only probit (in which case PCL is skipped and a standard probit GLM is fitted). Default is `"Z"`.
pair_type	Character string specifying which within-subject visit pairs to use for the pairwise composite likelihood. One of: `"adjacent"` (Default.) Use only temporally adjacent pairs $(j, j+1)$. This is computationally cheaper and typically sufficient. `"all"` Use all $\binom{m_i}{2}$ pairs per subject. More informative but $O(m_i^2)$ per subject.
phi_init	Numeric scalar giving the starting value for the OU decay parameter ϕ in the optimisation. Default is `0.1`.
phi_bounds	Numeric vector of length 2 giving the lower and upper bounds for ϕ (on the original scale; internally log-transformed). Default is `c(1e-4, 100)`.

`optim_method` Character string specifying the optimisation method passed to `optim`. Default is `"L-BFGS-B"`.

`verbose` Logical; if `TRUE`, progress messages are printed during estimation. Default is `FALSE`.

Details

The model extends `GIVEHR` by introducing an Ornstein–Uhlenbeck process $\xi_i(t)$ in the observation model:

$$\tilde{R}_{ij} = \alpha^\top X_{ij}^O + \delta^\top Z_{ij}^O \cdot U_i + q_i^\top Z_{ij}^O + \xi_i(t_{ij}),$$

where $R_{ij} = I(\tilde{R}_{ij} > 0)$, $q_i \sim N(0, \Sigma_q)$, and ξ_i is a stationary OU process with unit marginal variance and autocorrelation $\mathrm{Cor}(\xi_i(s), \xi_i(t)) = \exp(-\phi|t-s|)$. The OU component captures temporal serial correlation in the observation process that is not explained by covariates or the shared latent factor.

Stage 2 – Pairwise Composite Likelihood. Because the full likelihood involves high-dimensional integrals over ξ_i, estimation proceeds via pairwise composite likelihood. For each within-subject visit pair (j, k) the bivariate probit probability $P(R_{ij} = r_j, R_{ik} = r_k)$ is computed using `pbivnorm`, and the composite log-likelihood is

$$cl(\theta) = \sum_i \sum_{j<k} \log \Phi_2(d_j z_j,\, d_k z_k;\; d_j d_k \rho_{jk}),$$

where $d_j = 2R_{ij} - 1$, z_j is the standardised marginal probit argument, and ρ_{jk} combines the latent factor, random-effect, and OU contributions to the pairwise correlation. An analytic gradient is provided and the composite log-likelihood is maximised with `optim` using the specified `optim_method`.

Stages 1 and 3 are identical to `GIVEHR`, except that Stage 3 uses a static (non-filtered) version of $\omega_i(t)$ and $\kappa_i(t)$.

S3 methods `print`, `summary`, `coef`, `vcov`, and `confint` are available for objects of class `"GIVEHR_OU_PCL"`.

Value

An object of class `"GIVEHR_OU_PCL"`, which is a named list with the following components:

`visiting_process` A list with elements:

- `gamma_V` Named numeric vector of visiting-process regression coefficients.
- `Lambda0_hat` Named numeric vector of Breslow baseline hazard increments at each unique event time.
- `sigma2_zeta` Numeric scalar; estimated log-normal frailty variance on the log scale.
- `mu_Ui` Named numeric vector of empirical Bayes posterior means for U_i.
- `sigma2_Ui` Named numeric vector of empirical Bayes posterior variances for U_i.

`observation_process` A list with elements:

- `alpha_fixed` Named numeric vector of fixed-effect probit coefficients (including intercept).
- `delta_latent_link` (Present only when random effects are fitted.) Named numeric vector of latent-link coefficients δ.
- `Sigma_q` (Present only when random effects are fitted.) Named numeric vector of estimated random-effect variances $\sigma^2_{q_l}$.
- `phi_OU` (Present only when random effects are fitted.) Numeric scalar; estimated OU decay parameter ϕ. Larger values indicate faster decay of serial correlation.

`pcl_value` (Present only when random effects are fitted.) Numeric scalar; minimised negative pairwise composite log-likelihood.

`convergence` (Present only when random effects are fitted.) Integer convergence code from `optim`.

`longitudinal_outcome` A list with elements:

`beta_Y` Named numeric vector of outcome fixed-effect coefficients.

`theta_Y` Named numeric vector of outcome latent-link coefficients.

`se_beta` Named numeric vector of sandwich standard errors for `beta_Y`.

`vcov_robust` Numeric matrix; sandwich variance–covariance matrix for `c(beta_Y, theta_Y)`.

`model_info` A list with elements:

`n_subjects` Integer; number of unique subjects.

`n_visits` Integer; total number of visits.

`n_observed` Integer; total number of visits with `R = 1`.

`pair_type` Character string echoing the `pair_type` argument.

`median_gap` Numeric scalar; median time gap between consecutive visits within subjects.

`call` The matched call.

References

Yang, C.-H., Shi, X., and Mukherjee, B. (2026). "Joint Modeling of Longitudinal EHR Data with Shared Random Effects for Informative Visiting and Observation Processes." arXiv:2602.15374. doi:10.48550/arXiv.2602.15374.

See Also

`GIVEHR` for the base version without OU serial correlation; `GIVEHR_GH` for the Gauss–Hermite quadrature variant; `sim_data_gen` for simulating data compatible with this model;

Examples

```
dat <- sim_ehr_data[sim_ehr_data$id <= 80, ]
fit <- GIVEHR_OU_PCL(data = dat, pair_type = "adjacent")
print(fit)
```

`IIRR_balancing` *Inverse Intensity Rate Ratio (IIRR) Estimator with Balancing Weights*

Description

Estimates marginal regression coefficients for a longitudinal outcome using the balancing-weights approach of Yiu and Su (2024). The visiting-process intensity is modelled via a partial-likelihood score equation, and the outcome coefficients are obtained from weighted estimating equations using two alternative weight sets: (i) balancing weights that satisfy covariate-balance constraints in the spirit of entropy balancing, and (ii) conventional Maximum Likelihood Estimation (MLE) based inverse intensity weights.

A sensitivity parameter ϕ allows the analyst to explore departures from the assumption that the visiting process is fully captured by the observed covariates.

Usage

```
IIRR_balancing(
  data,
  id = "id",
  time = "time",
  outcome = "Y",
  obs_indicator = "R",
  visit_covs = c("Z", "X"),
  balance_covs = NULL,
  formula = NULL,
  phi = 0
)
```

Arguments

`data`	A `data.frame` in long (panel) format with one row per subject-time combination. Must contain columns for subject identifier, visit time, outcome, observation indicator, and all specified covariates.
`id`	Character string giving the name of the subject identifier column. Default is `"id"`.
`time`	Character string giving the name of the visit-time column. Default is `"time"`.
`outcome`	Character string giving the name of the outcome column. Default is `"Y"`.
`obs_indicator`	Character string giving the name of the binary observation indicator column (1 = outcome observed at visit). Default is `"R"`.
`visit_covs`	Character vector of covariate column names for the visiting-process intensity model. Default is `c("Z", "X")`.
`balance_covs`	Character vector of covariate column names for which covariate-balance constraints are imposed. If `NULL` (the default), `visit_covs` is used.
`formula`	An optional `formula` specifying the design matrix for the longitudinal outcome model. When `NULL` (the default), the formula is constructed automatically as `outcome ~ visit_covs + time` from the column-name arguments. A user-supplied formula overrides this (e.g., `Y ~ Z + X + Z:X + time` for interactions).
`phi`	Numeric scalar; sensitivity parameter controlling departures from the semiparametric visiting-process model. When $\phi = 0$ (the default), the estimator coincides with the standard IIRR assumption. Non-zero ϕ tilts the balancing weights to explore sensitivity to unmeasured confounding of the visit process.

Details

The estimator proceeds in four steps:

Step 1 – Visiting-Process Coefficients. The partial-likelihood score equation for the visiting-process coefficients γ is solved with `nleqslv`:

$$\sum_j \big[Z_{ij} - \bar{Z}(\gamma; t_j)\big] = 0.$$

Step 2 – MLE Weights. The MLE-based inverse intensity weight for observation (i, j) is

$$w_i^{\text{MLE}} = \frac{1}{\exp(\hat{\gamma}^\top Z_i)\hat{\Lambda}_0(C)},$$

where $\hat{\Lambda}_0(C)$ is the Breslow cumulative baseline hazard estimate.

Step 3 – Balancing Weights. The balancing weights w_i^{bal} are found by solving a constrained entropy-minimisation problem:

$$\min \sum_i w_i \log(w_i) \quad \text{s.t.} \quad \sum_i w_i X_i = \bar{X},$$

where $\bar{X}$ is the unweighted covariate mean and the constraint set is determined by `balance_covs`. When $\phi \neq 0$, the constraints are tilted via the sensitivity parameter. The dual problem is solved with `nleqslv` using Broyden's method.

Step 4 – Outcome Estimation. The outcome coefficients β are estimated by weighted least squares using both the balancing weights (`beta_bal`) and the MLE weights (`beta_mle`).

S3 methods `print` and `summary` are available for objects of class `"IIRR_balancing"`.

Value

An object of class `"IIRR_balancing"`, which is a named list with the following components:

`beta_bal` Named numeric vector of outcome regression coefficients estimated with the proposed balancing weights.

`beta_mle` Named numeric vector of outcome regression coefficients estimated with conventional MLE-based inverse intensity weights (for comparison).

`gamma_hat` Named numeric vector of visiting-process regression coefficients, estimated by solving the partial-likelihood score equation with `nleqslv`.

`balancing_weights` Numeric vector of per-observation balancing weights.

`mle_weights` Numeric vector of per-observation MLE-based inverse intensity weights.

`breslow` Numeric scalar; Breslow estimate of the cumulative baseline hazard $\Lambda_0(C)$.

`constraints` Numeric vector of constraint residuals (ideally near zero when balance is achieved).

`phi` Numeric scalar echoing the sensitivity parameter.

`convergence` A list with elements:

- `gamma` Integer; convergence code from `nleqslv` for the visiting-process estimation (1 = converged).
- `balancing` Integer; convergence code from the balancing-weights optimisation (1 = converged).

Note

This function requires the **nleqslv** package.

References

Sean Yiu and Li Su (2025). Accommodating informative visit times for analysing irregular longitudinal data: a sensitivity analysis approach with balancing weights estimators. *Journal of the Royal Statistical Society Series C: Applied Statistics*, 74(3), 824–843. doi:10.1093/jrsssc/qlaf002.

Examples

```
dat <- sim_ehr_data[sim_ehr_data$id <= 100, ]
fit <- IIRR_balancing(dat, visit_covs = c("Z", "X"))
print(fit)
```

`IIRR_weighting` *Inverse Intensity Rate Ratio (IIRR) Weighting Estimator*

Description

Estimates marginal regression coefficients for a longitudinal outcome by weighting each observation by the inverse of its estimated visit intensity. The visiting process is modelled via a Cox-type partial likelihood score equation, and the outcome coefficients are obtained by solving inverse-intensity-weighted estimating equations.

Only rows with observed outcomes (`R == 1` or `R` is `NA`) are used in both the visiting-process and outcome estimation.

Usage

```
IIRR_weighting(
  data,
  id = "id",
  time = "time",
  outcome = "Y",
  obs_indicator = "R",
  visit_covs = c("Z", "X"),
  formula = NULL
)
```

Arguments

`data`	A `data.frame` in long (panel) format with one row per subject-time combination. Must contain columns for subject identifier, visit time, outcome, observation indicator, and all specified covariates.
`id`	Character string giving the name of the subject identifier column. Default is `"id"`.
`time`	Character string giving the name of the visit-time column. Default is `"time"`.
`outcome`	Character string giving the name of the outcome column. Default is `"Y"`.
`obs_indicator`	Character string giving the name of the binary observation indicator column (1 = outcome observed at visit). Default is `"R"`.
`visit_covs`	Character vector of covariate column names for the visiting-process intensity model. These enter the exponent of the multiplicative intensity. Default is `c("Z", "X")`.
`formula`	An optional `formula` specifying the design matrix for the longitudinal outcome model. When `NULL` (the default), the formula is constructed automatically as `outcome ~ visit_covs + time` from the column-name arguments. A user-supplied formula overrides this (e.g., `Y ~ Z + X + Z:X + time` for interactions).

Details

The estimator proceeds in three steps:

Step 1. The visiting-process coefficients γ are estimated by solving the partial-likelihood score equation

$$\sum_j \big[Z_{ij} - \bar{Z}(\gamma; t_j)\big] = 0,$$

where $\bar{Z}(\gamma; t) = \sum_i Z_i \exp(\gamma^\top Z_i) / \sum_i \exp(\gamma^\top Z_i)$ is the risk-set mean.

Step 2. Inverse intensity weights are computed as $w_i = \exp(\hat{\gamma}^\top Z_i)$.

Step 3. The outcome coefficients β are estimated by solving the weighted estimating equation

$$\sum_{i,j} w_i^{-1} X_{ij} (Y_{ij} - X_{ij}^\top \beta) = 0.$$

Value

An object of class `"IIRR_weighting"`, which is a named list with two components:

`gamma_hat` Named numeric vector of visiting-process regression coefficients, estimated by solving the partial-likelihood score equation with `nleqslv`.

`beta_hat` Named numeric vector of outcome model coefficients, estimated from inverse-intensity-weighted estimating equations solved with `nleqslv`.

Note

This function requires the **nleqslv** package.

No standard errors are returned. For inference, consider using `IIRR_balancing` which provides balancing-weights estimates, or bootstrap the entire procedure.

References

Petra Bůžková and Thomas Lumley (2007). Longitudinal data analysis for generalized linear models with follow-up dependent on outcome-related variables. *The Canadian Journal of Statistics*, 35(4), 485–500. doi:10.1002/cjs.5550350402.

Examples

```
dat <- sim_ehr_data[sim_ehr_data$id <= 100, ]
fit <- IIRR_weighting(dat, visit_covs = c("Z", "X"))
print(fit)
```

`JMVL_Liang` *Joint Model for Visiting and Longitudinal (JMVL) Processes with Informative Presence (IP) Adjustment (Liang's Method)*

Description

Fits a joint model for longitudinal data with an informative visiting process using the estimating-equations approach of Liang, Lu, and Ying (2009). The visiting-process intensity is modelled via a Cox-type partial-likelihood score equation, and the longitudinal outcome is estimated by solving Liang-type risk-set-centred estimating equations that incorporate an empirical Bayes estimate of the shared latent factor.

Unlike `EHRJoint`, this method does not model a separate observation process; it assumes that the outcome is always observed at visits (or equivalently, filters to rows with `R == 1` or `R` is `NA`).

Usage

```
JMVL_Liang(
  data,
  id = "id",
  time = "time",
  outcome = "Y",
  obs_indicator = "R",
  visit_covs = c("Z", "X"),
  long_covs = c("Z", "X"),
  random_covs = "Z"
)
```

Arguments

data	A data.frame in long (panel) format with one row per subject-time combination. Must contain columns for subject identifier, visit time, outcome, observation indicator, and all specified covariates.
id	Character string giving the name of the subject identifier column. Default is "id".
time	Character string giving the name of the visit-time column. Default is "time".
outcome	Character string giving the name of the outcome column. Default is "Y".
obs_indicator	Character string giving the name of the binary observation indicator column. Rows with R == 1 or R is NA are retained; all others are dropped. Default is "R".
visit_covs	Character vector of covariate column names for the visiting-process intensity model. Default is c("Z", "X").
long_covs	Character vector of covariate column names for the fixed-effect part of the longitudinal outcome model. Default is c("Z", "X").
random_covs	Character vector of covariate column names that interact with the latent factor in the outcome model. An intercept column is prepended automatically. Default is "Z".

Details

The model consists of two linked components:

Visiting Process. The visit intensity is

$$\lambda_i(t) = \eta_i \exp(\gamma^\top Z_i)\, \lambda_0(t),$$

where η_i is a shared frailty. The coefficient vector γ is estimated by solving the partial-likelihood score equation, the cumulative baseline hazard $\Lambda_0(C)$ by the Breslow estimator, and the frailty variance σ^2 by method of moments.

Outcome Model. An empirical Bayes estimator of the latent loading B_i is constructed as

$$\hat{B}_i = \frac{1 + n_i \hat{\sigma}^2}{1 + \hat{\Lambda}_i \hat{\sigma}^2} - 1,$$

and the outcome model

$$Y_i(t) = \beta^\top X_i(t) + B_i Z_i^*(t) + \varepsilon_i(t)$$

is estimated by solving risk-set-centred estimating equations

$$\sum_{i,j} \big[(M_{ij} - \bar{M}(t_j))\big](Y_{ij} - M_{ij}^\top \beta^*) = 0,$$

where $M_{ij} = (X_{ij}, \hat{B}_i Z_{ij}^*)$ and $\bar{M}(t)$ is the risk-set-weighted mean.

This differs from EHRJoint in that no observation-process weights enter the estimating equations.

Value

A named list with four components:

`gamma_hat` Named numeric vector of visiting-process regression coefficients, estimated by solving the partial-likelihood score equation with `nleqslv`.

`Lambda_C_est` Numeric scalar; Breslow estimate of the cumulative baseline hazard $\Lambda_0(C)$ over the full follow-up window.

`sigma_sq_hat` Numeric scalar; method-of-moments estimate of the frailty variance σ^2. Constrained to be non-negative.

`beta_hat` Named numeric vector of outcome fixed-effect coefficients, estimated from the Liang-type estimating equations.

Note

This function requires the **nleqslv** package. It will attempt to install the package automatically if it is not available.

No standard errors are returned. For inference, consider bootstrapping the entire procedure.

References

Yu Liang, Wenbin Lu, and Zhiliang Ying (2009). Joint modeling and analysis of longitudinal data with informative observation times. *Biometrics*, 65(2), 377–384. doi:10.1111/j.15410420.2008.01104.x.

Examples

```
dat <- sim_ehr_data[sim_ehr_data$id <= 80, ]
fit <- JMVL_Liang(dat)
print(fit)
```

`LI_IP` *Linear Increment (LI) Imputation with Informative Presence (IP) Adjustment*

Description

Implements a two-stage procedure that first imputes missing outcome values using the linear increment (LI) model of Aalen and Gunnes (2010), then applies an informative-presence (IP) adjustment method to the completed dataset. The LI model is fitted with `slim` and handles irregular observation times naturally by modelling increments rather than levels.

Unlike `MI_IP`, this approach produces a single completed dataset (no pooling required), making it computationally faster but unable to account for imputation uncertainty in standard errors.

Usage

```
LI_IP(
  data,
  id = "id",
  time = "time",
  outcome = "Y",
  obs_indicator = "R",
  li_formula = NULL,
  li_covariance = c("randomwalk", "independence", "exchangeable"),
  li_limit = ~1,
  ip_method = c("LMM", "Liang", "IIRR_weighting", "IIRR_balancing", "PairLik"),
  visit_covs = c("Z", "X"),
  outcome_covs = NULL,
  random_covs = NULL,
  outcome_formula = NULL,
  random_formula = NULL,
  ip_args = list()
)
```

Arguments

data	A `data.frame` in long (panel) format with one row per subject-time combination. Must contain columns for subject identifier, visit time, outcome, observation indicator, and all specified covariates. Rows with `NA` in the time column are treated as unvisited time points.
id	Character string giving the name of the subject identifier column. Default is `"id"`.
time	Character string giving the name of the visit-time column. Default is `"time"`.
outcome	Character string giving the name of the outcome column. Default is `"Y"`.
obs_indicator	Character string giving the name of the binary observation indicator column (`1` = outcome observed, `0` = outcome missing at visit). Default is `"R"`.
li_formula	A `formula` for the linear increment model. The left-hand side should be the outcome variable. Default is `NULL`, which constructs `Y ~ <visit_covs> + <time>`.
li_covariance	Character string specifying the covariance structure for the linear increment model. One of: `"randomwalk"` (Default.) Random-walk covariance, recommended for irregular data. The variance of increments is proportional to the time gap. `"independence"` Independent increments. `"exchangeable"` Exchangeable correlation structure.
li_limit	A one-sided `formula` for the limit specification passed to `slim`. Default is `~ 1` (constant limit).
ip_method	Character string specifying the IP adjustment method applied to the completed dataset. One of: `"LMM"` Standard linear mixed model via `LMM` (no IP adjustment; included for comparison). `"Liang"` Joint model of Liang et al. (2009) via `JMVL_Liang`. `"IIRR_weighting"` Inverse intensity rate ratio weighting via `IIRR_balancing` (using MLE weights).

"IIRR_balancing" Balancing-weights estimator of Yiu and Su (2024) via IIRR_balancing.

"PairLik" Pairwise composite likelihood via PairwiseLikelihood.

visit_covs Character vector of covariate column names for the visiting-process model (used by Liang and IIRR methods). Default is c("Z", "X").

outcome_covs Character vector of covariate column names for the outcome model in the Liang method. Default is NULL (uses visit_covs).

random_covs Character vector of covariate column names with random effects in the Liang method. Default is NULL (uses the first element of visit_covs).

outcome_formula

A formula for the longitudinal outcome model, used by LMM, IIRR, and PairLik methods. Default is NULL, which constructs Y ~ <outcome_covs> + <time>.

random_formula A one-sided formula for the LMM random effects (e.g., ~ 1 | id). Default is NULL, which constructs ~ 1 | <id>.

ip_args A named list of additional arguments passed to the IP method. Common options include:

For IIRR methods: phi (numeric; sensitivity parameter, default 0).

For PairLik: pair_sample (integer; number of pairs to sample).

For LMM: visit_adjust ("OA" or "VA").

Details

The procedure has two stages:

Stage 1 — Linear Increment Imputation. The linear increment model (Aalen and Gunnes, 2010) models the change in outcome between consecutive observations:

$$\Delta Y_{ij} = (t_j - t_{j-1})(\beta_0 + \beta^\top X_{ij}) + \varepsilon_{ij},$$

where $\mathrm{Var}(\varepsilon_{ij}) \propto (t_j - t_{j-1})$ under the random-walk covariance structure. The model is fitted with slim on the subset of rows with observed outcomes, and predictions are generated for rows where the outcome is missing but visit time is known.

Stage 2 — IP Adjustment. The completed dataset is passed to the selected IP method for coefficient estimation. Because a single imputed dataset is used (rather than multiple imputations), the reported standard errors (when available) do not account for imputation uncertainty.

S3 methods print and summary are available for objects of class "LI_IP_result".

Value

An object of class "LI_IP_result", which is a named list with the following components:

coefficients Named numeric vector of estimated regression coefficients from the IP method.

se Named numeric vector of standard errors (when available from the IP method; NULL otherwise).

ip_method Character string echoing the IP method used.

ip_args List echoing the additional IP arguments.

li_model A list with elements formula, covariance, and limit describing the LI imputation model.

n_imputed Integer; number of missing outcome values that were imputed.

imputed_data A data.frame; the completed dataset after LI imputation.

call The matched call.

References

Odd O. Aalen and Nina Gunnes (2010). A dynamic approach for reconstructing missing longitudinal data using the linear increments model. *Biostatistics*, 11(3), 453–472. doi:10.1093/biostatistics/kxq014.

Examples

```
dat <- sim_ehr_data[sim_ehr_data$id <= 80,
                    c("id", "Z", "X", "time", "R", "Y", "C")]
fit <- LI_IP(dat, ip_method = "LMM", visit_covs = c("Z", "X"))
print(fit)
```

LMM *Fit a Linear Mixed Model (LMM) for Longitudinal Data*

Description

Fits a linear mixed-effects model (LMM) to longitudinal data using `lme`, with optional filtering by an observation indicator and optional visit-count adjustment. This provides a convenient wrapper for standard LMM, Observation-Adjusted LMM (OA-LMM), and Visit-Adjusted LMM (VA-LMM) estimators commonly used as benchmarks in informative-visiting analyses.

Usage

```
LMM(
  data,
  id = "id",
  fixed = Y ~ 1,
  random = ~1 | id,
  obs_indicator = NULL,
  visit_adjust = NULL,
  method = "REML",
  optimizer = "nlminb"
)
```

Arguments

`data`	A `data.frame` in long (panel) format containing longitudinal data with one row per subject-time combination.
`id`	Character string giving the name of the subject identifier column. This is used for the visit-count adjustment (if requested) and is referenced in the random-effects formula. Default is `"id"`.
`fixed`	A `formula` specifying the fixed-effects structure (e.g., `Y ~ Z + X + time`). When `visit_adjust` is non-`NULL`, the term `n_visits` is appended automatically. Default is `Y ~ 1`.
`random`	A one-sided `formula` specifying the random-effects structure using **nlme** syntax (e.g., `~ 1 + Z | id`). Default is `~ 1 | id`.
`obs_indicator`	Character string giving the name of the observation indicator column. Rows where this column equals `0` are dropped before fitting. Set to `NULL` to use all rows. Default is `NULL`.

`visit_adjust` Character string specifying the type of visit-count adjustment. One of:

`NULL` (Default.) No adjustment; a standard LMM is fitted.

`"OA"` Observed-adjusted: the cumulative visit count `n_visits` is computed *after* filtering by `obs_indicator`. This adjusts for the number of observed (retained) visits.

`"VA"` Visit-adjusted: the cumulative visit count `n_visits` is computed *before* filtering by `obs_indicator`. This adjusts for the total number of visits regardless of whether the outcome was observed.

When either `"OA"` or `"VA"` is specified, the variable `n_visits` is automatically added to the `fixed` formula.

`method` Character string specifying the estimation method passed to `lme`. One of `"REML"` (restricted maximum likelihood; default) or `"ML"` (maximum likelihood).

`optimizer` Character string specifying the optimiser for the `lmeControl` object. One of `"nlminb"` (default) or `"optim"`.

Details

This function is a convenience wrapper that:

1. Optionally computes a within-subject cumulative visit counter (`n_visits`) either before or after filtering by the observation indicator.
2. Filters out rows where `obs_indicator == 0` (if specified).
3. Removes incomplete cases.
4. Fits a linear mixed model with `lme` using the specified fixed and random formulas, estimation method, and optimiser.

The OA-LMM and VA-LMM adjustments were proposed as simple corrections for informative visit times. OA-LMM conditions on the number of visits that yielded observed outcomes, while VA-LMM conditions on the total number of visits. Neither fully accounts for the visiting process but can reduce bias compared to a naive LMM.

Value

A numeric matrix of fixed-effect estimates, identical to the `tTable` component of `summary(lme(...))`. The matrix has one row per fixed effect and columns `Value`, `Std.Error`, `DF`, `t-value`, and `p-value`.

References

Benjamin A. Goldstein, Nrupen A. Bhavsar, Matthew Phelan, and Michael J. Pencina (2016). Controlling for informed presence bias due to the number of health encounters in an electronic health record. *American Journal of Epidemiology*, 184(11), 847–855. doi:10.1093/aje/kww112.

Nan M. Laird and James H. Ware (1982). Random-effects models for longitudinal data. *Biometrics*, 38(4), 963–974.

Examples

```
dat <- sim_ehr_data[sim_ehr_data$id <= 100, ]
fit <- LMM(dat, fixed = Y ~ Z + X + time,
           random = ~ 1 | id, obs_indicator = "R")
print(fit)
```

MI_IP *Multiple Imputation (MI) with Informative Presence (IP) Adjustment*

Description

Implements a two-stage procedure that first imputes missing outcome values using a multilevel model via mice, then applies an informative-presence (IP) adjustment method to each imputed dataset and pools results. The imputation model accounts for the clustered structure of longitudinal data using a two-level specification (random intercept and optional random slopes).

Unlike LI_IP, this approach generates m completed datasets and pools estimates, allowing imputation uncertainty to be reflected in the standard errors (when Rubin's rules are used).

Usage

```
MI_IP(
  data,
  id = "id",
  time = "time",
  outcome = "Y",
  obs_indicator = "R",
  m = 5,
  maxit = 50,
  impute_method = c("2l.pan", "2l.lmer", "2l.norm"),
  impute_fixed = NULL,
  impute_random = NULL,
  impute_include_R = TRUE,
  ip_method = c("LMM", "Liang", "IIRR_weighting", "IIRR_balancing", "PairLik"),
  visit_covs = c("Z", "X"),
  outcome_covs = NULL,
  random_covs = NULL,
  outcome_formula = NULL,
  random_formula = NULL,
  ip_args = list(),
  pool_method = c("mean", "rubin")
)
```

Arguments

data	A data.frame in long (panel) format with one row per subject-time combination. Must contain columns for subject identifier, visit time, outcome, observation indicator, and all specified covariates. Rows with NA in the time column are treated as unvisited time points.
id	Character string giving the name of the subject identifier column. Default is "id".
time	Character string giving the name of the visit-time column. Default is "time".
outcome	Character string giving the name of the outcome column. Default is "Y".
obs_indicator	Character string giving the name of the binary observation indicator column (1 = outcome observed, 0 = outcome missing at visit). Default is "R".
m	Integer; number of multiply imputed datasets. Default is 5.

`maxit` Integer; maximum number of iterations for `mice`. Default is `50`.

`impute_method` Character string specifying the **mice** imputation method for the outcome. One of:

`"2l.pan"` (Default.) Two-level model with heterogeneous within-group variance, implemented via **pan**.

`"2l.lmer"` Two-level model using `lmer`.

`"2l.norm"` Two-level normal model.

`impute_fixed` Character vector of covariate column names to include as fixed effects in the imputation model. Default is `NULL`, which uses `c(visit_covs, time)`. The imputation model for Y is:

$$Y_{ij} = \beta_0 + \sum_k \beta_k X_{kij} + b_{0i} + \sum_l b_{li} Z_{lij} + \varepsilon_{ij},$$

where fixed effects include an intercept plus all variables in `impute_fixed`.

`impute_random` Character vector of covariate column names to include as random slopes in the imputation model. Default is `NULL` (random intercept only). For example, `c("Z")` adds a random slope for Z: $b_{0i} + b_{1i} Z_{ij}$.

`impute_include_R` Logical; whether to include the observation indicator `R` as a predictor in the imputation model. Default is `TRUE`. Recommended because `R` may be informative about missing Y values under MAR.

`ip_method` Character string specifying the IP adjustment method applied to each completed dataset. One of:

`"LMM"` Standard linear mixed model via `LMM` (no IP adjustment; for comparison).

`"Liang"` Joint model of Liang et al. (2009) via `JMVL_Liang`.

`"IIRR_weighting"` Inverse intensity rate ratio weighting via `IIRR_balancing` (MLE weights).

`"IIRR_balancing"` Balancing-weights estimator of Yiu and Su (2024) via `IIRR_balancing`.

`"PairLik"` Pairwise composite likelihood via `PairwiseLikelihood`.

`visit_covs` Character vector of covariate column names for the visiting-process model (used by Liang and IIRR methods). Default is `c("Z", "X")`.

`outcome_covs` Character vector of covariate column names for the outcome model in the Liang method. Default is `NULL` (uses `visit_covs`).

`random_covs` Character vector of covariate column names with random effects in the Liang method. Default is `NULL` (uses the first element of `visit_covs`).

`outcome_formula` A `formula` for the longitudinal outcome model, used by LMM, IIRR, and PairLik methods. Default is `NULL`, which constructs `Y ~ <outcome_covs> + <time>`.

`random_formula` A one-sided `formula` for the LMM random effects. Default is `NULL`, which constructs `~ 1 | <id>`.

`ip_args` A named list of additional arguments passed to the IP method. Common options include:

For IIRR methods: `phi` (numeric; sensitivity parameter, default `0`). `balance_covs` (character vector).

For PairLik: `pair_sample` (integer; number of pairs to sample). `family` (`"gaussian"` or `"binomial"`).

For LMM: `method` (`"REML"` or `"ML"`). `optimizer` (`"nlminb"` or `"optim"`). `visit_adjust` (`NULL`, `"OA"`, or `"VA"`).

`pool_method` Character string specifying the method for pooling estimates across imputations. One of:

`"mean"` (Default.) Simple averaging of point estimates with Monte Carlo standard errors.

`"rubin"` Rubin's rules: combines within-imputation and between-imputation variance for valid inference. Requires that the IP method returns standard errors.

Details

The procedure has three stages:

Stage 1 — Multiple Imputation. Missing outcome values (where `R = 0` but time is not `NA`) are imputed using a multilevel model via `mice`. Unvisited rows (time is `NA`) are separated, imputation is performed on the visited subset, and unvisited rows are reattached afterwards. The predictor matrix is configured so that only the outcome is imputed and only the specified covariates are used as predictors.

Stage 2 — IP-Adjusted Analysis. Each completed dataset is analysed by the selected IP method. Failed fits are discarded with a warning.

Stage 3 — Pooling. Point estimates are combined across successful imputations. Under `pool_method = "rubin"`, Rubin's rules are applied: the total variance is $T = \bar{W} + (1 + 1/m)B$, where $\bar{W}$ is the mean within-imputation variance and B is the between-imputation variance.

S3 methods `print` are available for objects of class `"MI_IP_result"`.

Value

An object of class `"MI_IP_result"`, which is a named list with the following components:

`coefficients` Named numeric vector of pooled regression coefficients.

`se` Named numeric vector of pooled standard errors.

`by_imputation` List of length `m`, each element a list with `coef` and (optionally) `se` from one imputed dataset.

`ip_method` Character string echoing the IP method used.

`ip_args` List echoing the additional IP arguments.

`impute_method` Character string echoing the imputation method.

`impute_model` A list with elements `fixed`, `random`, and `include_R` describing the imputation model.

`m` Integer; number of successful imputations.

`m_requested` Integer; number of imputations originally requested.

`call` The matched call.

References

Donald B. Rubin (1987). *Multiple Imputation for Nonresponse in Surveys*. John Wiley & Sons. doi:10.1002/9780470316696.

Examples

```
dat <- sim_ehr_data[sim_ehr_data$id <= 80,
                    c("id", "Z", "X", "time", "R", "Y", "C")]
fit <- MI_IP(dat, ip_method = "LMM",
             visit_covs = c("Z", "X"), m = 2, maxit = 5)
print(fit)
```

`PairwiseLikelihood` *Pairwise Composite Likelihood (PCL) Generalized Linear Model (GLM) Estimation*

Description

Fits a Generalized Linear Model (GLM) using Pairwise Composite Likelihood (PCL). For each pair of subjects, the contribution to the composite log-likelihood is based on the difference $Y_i - Y_j$ (Gaussian family) or a logistic-style softplus function (binomial family). This approach yields consistent slope estimates without requiring a correctly specified within-cluster correlation structure, and is robust to informative cluster sizes.

The intercept is not identifiable in pairwise differences and is automatically removed if present in the formula.

Usage

```
PairwiseLikelihood(
  data,
  formula,
  id_col = "id",
  time_col = "time",
  y_col = "Y",
  family = c("gaussian", "binomial"),
  start = NULL,
  pair_sample = NULL,
  pair_seed = 1L,
  hessian = TRUE,
  verbose = FALSE
)
```

Arguments

`data`	A `data.frame` containing longitudinal or clustered data. Rows with missing values in the outcome, covariates, or identifier columns are removed.
`formula`	A `formula` specifying the model (e.g., `Y ~ Z + X + time`). Must be supplied by the user; there is no default. If an intercept is present it is removed automatically with a message (when `verbose = TRUE`).
`id_col`	Character string giving the name of the subject/cluster identifier column. Default is `"id"`.
`time_col`	Character string giving the name of the time variable column. Default is `"time"`.
`y_col`	Character string giving the name of the outcome column. Default is `"Y"`.

`family` Character string specifying the response distribution. One of:

`"gaussian"` (Default.) Pairwise residual sum of squares: $\sum 0.5\,(Y_i - Y_j - \mu_i + \mu_j)^2$.

`"binomial"` Logistic-style softplus: $\sum -\log(1 + \exp((Y_i - Y_j)(\mu_j - \mu_i)))$. Requires $Y \in \{0, 1\}$.

`start` Optional numeric vector of starting values for the regression coefficients. Default is `NULL`, which uses Ordinary Least Squares (Gaussian) or GLM (binomial) estimates.

`pair_sample` Optional positive integer; if the total number of subject pairs exceeds this value, a random subsample of pairs is used. Default is `NULL` (use all pairs).

`pair_seed` Integer seed for reproducible pair subsampling. Default is `1`.

`hessian` Logical; if `TRUE`, the numeric Hessian is computed at the optimum via `hessian` and used to obtain a naive variance–covariance matrix. Default is `TRUE`.

`verbose` Logical; if `TRUE`, progress messages are printed. Default is `FALSE`.

Details

The pairwise composite likelihood objective aggregates contributions from all (or a subsample of) between-subject pairs. For each pair of subjects (i, j) and all cross-products of their observations, the contribution is:

Gaussian:

$$l_{ij}(\beta) = -\frac{1}{2} \sum_{s \in i} \sum_{t \in j} \big[(Y_s - Y_t) - (\mu_s - \mu_t)\big]^2,$$

where $\mu_s = X_s^\top \beta$. Minimising the total gives the PCL estimator.

Binomial:

$$l_{ij}(\beta) = \sum_{s \in i} \sum_{t \in j} -\log\big[1 + \exp\big((Y_s - Y_t)(\mu_t - \mu_s)\big)\big].$$

The inner loops are implemented in C++ via **Rcpp** for efficiency. Optimization is performed with `optim` using BFGS.

S3 methods `print` and `summary` are available for objects of class `"PairwiseLikelihood"`.

Value

An object of class `"PairwiseLikelihood"`, which is a named list with the following components:

`call` The matched call.

`family` Character string echoing the response family.

`formula` The formula used (intercept removed if applicable).

`coef` Named numeric vector of estimated regression coefficients.

`loglik` Numeric scalar; composite log-likelihood at the optimum (Gaussian: negated pairwise residual sum of squares; binomial: pairwise softplus sum).

`converged` Logical; `TRUE` if `optim` converged.

`hessian` (When `hessian = TRUE`.) Numeric matrix; Hessian of the objective at the optimum.

`vcov_naive` (When `hessian = TRUE`.) Numeric matrix; inverse Hessian, a naive (model-based) variance–covariance estimator for `coef`.

`n` Integer; number of observations used.

`p` Integer; number of regression coefficients.

`G` Integer; number of unique subjects/clusters.

`k_pairs` Integer; number of subject pairs used.

`pair_sample` Integer or `NULL`; echoing the subsampling parameter.

Note

This function requires the **Rcpp** and **numDeriv** packages. The inline C++ functions `pw_total_gaussian_rcpp` and `pw_total_softplus_rcpp` are compiled on first use.

References

Weining Shen, Suyu Liu, Yong Chen, and Jing Ning (2019). Regression analysis of longitudinal data with outcome-dependent sampling and informative censoring. *Scandinavian Journal of Statistics*, 46(3), 831–847. doi:doi/10.1111/sjos.12373.

Yong Chen, Jing Ning, and Chunyan Cai (2015). Regression analysis of longitudinal data with irregular and informative observation times. *Biostatistics*, 16(4), 727–739. doi:10.1093/biostatistics/kxv008.

Examples

```
dat <- sim_ehr_data[sim_ehr_data$id <= 100 &
                    sim_ehr_data$R == 1 & !is.na(sim_ehr_data$R), ]
fit <- PairwiseLikelihood(dat, formula = Y ~ Z + X + time,
                          pair_sample = 200)
print(fit)
```

---

print.cimehr_bootstrap

Print a Bootstrap Result

Description

Prints bootstrap standard errors and percentile confidence intervals for outcome coefficients.

Usage

```
## S3 method for class 'cimehr_bootstrap'
print(x, digits = 4, ...)
```

Arguments

`x`	A `cimehr_bootstrap` object from `bootstrap`.
`digits`	Number of significant digits. Default is 4.
`...`	Ignored.

Value

`x`, invisibly.

See Also

summary.cimehr_bootstrap, bootstrap

Examples

```
dat <- sim_ehr_data[sim_ehr_data$id <= 100, ]
fit <- IIRR_weighting(dat, visit_covs = c("Z", "X"))
bs  <- bootstrap(fit, data = dat, B = 5, seed = 1)
print(bs)
```

---

print.EHRJoint *Print an EHRJoint Fit*

Description

Prints outcome fixed-effect estimates from the joint model without a separate observation process.

Usage

```
## S3 method for class 'EHRJoint'
print(x, digits = 4, level = 0.95, ...)
```

Arguments

x	An EHRJoint object returned by EHRJoint.
digits	Number of significant digits. Default is 4.
level	Confidence level for Wald intervals. Default is 0.95.
...	Ignored.

Value

x, invisibly.

See Also

summary.EHRJoint, EHRJoint

Examples

```
dat <- sim_ehr_data[sim_ehr_data$id <= 80, ]
fit <- EHRJoint(dat)
print(fit)
```

print.GIVEHR *Print a GIVEHR Fit*

Description

Prints outcome fixed-effect estimates with sandwich standard errors and Wald confidence intervals.

Usage

```
## S3 method for class 'GIVEHR'
print(x, digits = 4, level = 0.95, ...)
```

Arguments

x	A GIVEHR object returned by GIVEHR.
digits	Number of significant digits. Default is 4.
level	Confidence level for Wald intervals. Default is 0.95.
...	Ignored.

Value

x, invisibly.

See Also

summary.GIVEHR, GIVEHR

Examples

```
dat <- sim_ehr_data[sim_ehr_data$id <= 80, ]
fit <- GIVEHR(data = dat)
print(fit)
```

print.GIVEHR_GH *Print a GIVEHR_GH Fit*

Description

Prints outcome fixed-effect estimates from the Gauss–Hermite quadrature variant.

Usage

```
## S3 method for class 'GIVEHR_GH'
print(x, digits = 4, level = 0.95, ...)
```

Arguments

x	A GIVEHR_GH object returned by GIVEHR_GH.
digits	Number of significant digits. Default is 4.
level	Confidence level for Wald intervals. Default is 0.95.
...	Ignored.

Value

x, invisibly.

See Also

summary.GIVEHR_GH, GIVEHR_GH

Examples

```
dat <- sim_ehr_data[sim_ehr_data$id <= 50, ]
fit <- GIVEHR_GH(data = dat, gh_points_U = 3, gh_points_q = 3)
print(fit)
```

---

print.GIVEHR_OU_PCL *Print a GIVEHR_OU_PCL Fit*

Description

Prints outcome fixed-effect estimates with robust standard errors from the OU pairwise composite likelihood variant.

Usage

```
## S3 method for class 'GIVEHR_OU_PCL'
print(x, digits = 4, level = 0.95, ...)
```

Arguments

x	A GIVEHR_OU_PCL object returned by GIVEHR_OU_PCL.
digits	Number of significant digits. Default is 4.
level	Confidence level for Wald intervals. Default is 0.95.
...	Ignored.

Value

x, invisibly.

See Also

summary.GIVEHR_OU_PCL, GIVEHR_OU_PCL

Examples

```
dat <- sim_ehr_data[sim_ehr_data$id <= 80, ]
fit <- GIVEHR_OU_PCL(data = dat, pair_type = "adjacent")
print(fit)
```

`print.IIRR_balancing` *Print an IIRR_balancing Fit*

Description

Prints outcome estimates from both balancing and MLE weights.

Usage

```
## S3 method for class 'IIRR_balancing'
print(x, digits = 4, level = 0.95, ...)
```

Arguments

`x`	An `IIRR_balancing` object.
`digits`	Number of significant digits. Default is 4.
`level`	Confidence level for Wald intervals. Default is 0.95.
`...`	Ignored.

Value

`x`, invisibly.

See Also

`summary.IIRR_balancing`, `IIRR_balancing`

Examples

```
dat <- sim_ehr_data[sim_ehr_data$id <= 100, ]
fit <- IIRR_balancing(dat, visit_covs = c("Z", "X"))
print(fit)
```

print.IIRR_weighting *Print an IIRR_weighting Fit*

Description

Prints outcome fixed-effect estimates from the inverse intensity rate ratio weighting estimator.

Usage

```
## S3 method for class 'IIRR_weighting'
print(x, digits = 4, level = 0.95, ...)
```

Arguments

x	An IIRR_weighting object.
digits	Number of significant digits. Default is 4.
level	Confidence level for Wald intervals. Default is 0.95.
...	Ignored.

Value

x, invisibly.

See Also

summary.IIRR_weighting, IIRR_weighting

Examples

```
dat <- sim_ehr_data[sim_ehr_data$id <= 100, ]
fit <- IIRR_weighting(dat, visit_covs = c("Z", "X"))
print(fit)
```

print.JMVL_Liang *Print a JMVL_Liang Fit*

Description

Prints outcome fixed-effect estimates from the Liang–Lu–Ying joint estimating equations approach.

Usage

```
## S3 method for class 'JMVL_Liang'
print(x, digits = 4, level = 0.95, ...)
```

Arguments

x	A JMVL_Liang object.
digits	Number of significant digits. Default is 4.
level	Confidence level for Wald intervals. Default is 0.95.
...	Ignored.

Value

x, invisibly.

See Also

summary.JMVL_Liang, JMVL_Liang

Examples

```
dat <- sim_ehr_data[sim_ehr_data$id <= 80, ]
fit <- JMVL_Liang(dat)
print(fit)
```

---

print.LI_IP_result *Print an LI_IP Fit*

Description

Prints outcome coefficients from the linear increment imputation with informative presence adjustment.

Usage

```
## S3 method for class 'LI_IP_result'
print(x, digits = 4, level = 0.95, ...)
```

Arguments

x	An LI_IP object returned by LI_IP.
digits	Number of significant digits. Default is 4.
level	Confidence level for Wald intervals. Default is 0.95.
...	Ignored.

Value

x, invisibly.

See Also

summary.LI_IP_result, LI_IP

Examples

```
dat <- sim_ehr_data[sim_ehr_data$id <= 80,
                    c("id", "Z", "X", "time", "R", "Y", "C")]
fit <- LI_IP(data = dat, ip_method = "LMM",
             visit_covs = c("Z", "X"))
print(fit)
```

`print.LMM_result` *Print an LMM Result*

Description

Prints the fixed-effects table from a linear mixed model fit.

Usage

```
## S3 method for class 'LMM_result'
print(x, digits = 4, level = 0.95, ...)
```

Arguments

`x`	An `LMM_result` object returned by `LMM`.
`digits`	Number of significant digits. Default is 4.
`level`	Confidence level for Wald intervals. Default is 0.95.
`...`	Ignored.

Value

`x`, invisibly.

See Also

`summary.LMM_result`, `LMM`

Examples

```
dat <- sim_ehr_data[sim_ehr_data$id <= 100, ]
fit <- LMM(dat, fixed = Y ~ Z + X + time,
           random = ~ 1 | id, obs_indicator = "R")
print(fit)
```

print.MI_IP_result *Print an MI_IP Fit*

Description

Prints pooled outcome coefficients from the multiple imputation with informative presence adjustment.

Usage

```
## S3 method for class 'MI_IP_result'
print(x, digits = 4, level = 0.95, ...)
```

Arguments

x	An MI_IP object returned by MI_IP.
digits	Number of significant digits. Default is 4.
level	Confidence level for Wald intervals. Default is 0.95.
...	Ignored.

Value

x, invisibly.

See Also

summary.MI_IP_result, MI_IP

Examples

```
dat <- sim_ehr_data[sim_ehr_data$id <= 80,
                    c("id", "Z", "X", "time", "R", "Y", "C")]
fit <- MI_IP(data = dat, ip_method = "LMM",
             visit_covs = c("Z", "X"), m = 2, maxit = 5)
print(fit)
```

print.PairwiseLikelihood *Print a PairwiseLikelihood Fit*

Description

Prints coefficients with naive standard errors (from the inverse Hessian) when available.

Usage

```
## S3 method for class 'PairwiseLikelihood'
print(x, digits = 4, level = 0.95, ...)
```

Arguments

x	A PairwiseLikelihood object returned by PairwiseLikelihood.
digits	Number of significant digits. Default is 4.
level	Confidence level for Wald intervals. Default is 0.95.
...	Ignored.

Value

x, invisibly.

See Also

summary.PairwiseLikelihood, PairwiseLikelihood

Examples

```
dat <- sim_ehr_data[sim_ehr_data$id <= 100 &
                    sim_ehr_data$R == 1 & !is.na(sim_ehr_data$R), ]
fit <- PairwiseLikelihood(dat, formula = Y ~ Z + X + time,
                          pair_sample = 200)
print(fit)
```

---

print.SummaryStat_result

Print a SummaryStat Result

Description

Prints coefficients from the summary-statistic regression.

Usage

```
## S3 method for class 'SummaryStat_result'
print(x, digits = 4, ...)
```

Arguments

x	A SummaryStat_result object returned by SummaryStat.
digits	Number of significant digits. Default is 4.
...	Ignored.

Value

x, invisibly.

See Also

summary.SummaryStat_result, SummaryStat

Examples

```
dat <- sim_ehr_data[sim_ehr_data$id <= 100, ]
fit <- SummaryStat(dat, formula = ~ Z + X)
print(fit)
```

`sim_data_gen` *Simulate Longitudinal Electronic Health Record (EHR) Data with Informative Visiting and Observation Processes*

Description

Generates synthetic longitudinal data from a joint model that includes an informative visiting process (Poisson with log-normal frailty), an informative observation process (probit with latent-factor-linked random effects and an Ornstein–Uhlenbeck (OU) serial correlation component), and a linear longitudinal outcome model with latent-factor-linked random effects. The data-generating mechanism matches the model specification in Yang, Shi, and Mukherjee (2025, Web Appendix S9) and is suitable for evaluating `GIVEHR`, `GIVEHR_GH`, and `GIVEHR_OU_PCL`.

Usage

```
sim_data_gen(
  n,
  time.start = 0,
  time.end = 60,
  seed = 123,
  pZ = 0.5,
  muX = 0,
  sdX = 1,
  gamma = c(intercept = -2.2, Z = 1, X = 1),
  sigma_zeta = 1,
  alpha = c(intercept = 1.5, Z = -0.5, X = -0.3),
  delta = c(intercept = -0.3, Z = -0.5),
  sigma_q = c(intercept = 0.3, Z = 0.2),
  phi = 0.1,
  beta = c(intercept = -2, Z = -0.5, X = 0.5),
  beta_t = 0.1,
  theta = c(intercept = 0.5, Z = 0.3),
  Sigma_b = matrix(c(1, 0, 0, 2), 2, 2),
  sigma_y = 1
)
```

Arguments

`n`	Integer; number of subjects to simulate.
`time.start`	Numeric scalar; start of the observation window. Default is `0`.
`time.end`	Numeric scalar; end of the observation window (censoring time for all subjects). Default is `60`.
`seed`	Integer; random seed for reproducibility. Default is `123`.

pZ	Numeric scalar in $[0, 1]$; probability that the binary covariate Z_i equals 1 (Bernoulli). Default is 0.5.
muX	Numeric scalar; mean of the continuous covariate $X_i \sim N(\mu_X, \sigma_X^2)$. Default is 0.
sdX	Positive numeric scalar; standard deviation of X_i. Default is 1.
gamma	Named numeric vector of length 3 with elements intercept, Z, and X giving the visiting-process log-intensity coefficients. Default is c(intercept = -2.2, Z = 1, X = 1).
sigma_zeta	Positive numeric scalar; standard deviation of the log-normal frailty on the log scale. Controls the degree of overdispersion in visit counts. Default is 1.0.
alpha	Named numeric vector with elements intercept, Z, and X giving the observation-process fixed-effect probit coefficients. Default is c(intercept = 1.5, Z = -0.5, X = -0.3).
delta	Named numeric vector with elements intercept and Z giving the latent-link coefficients that couple the shared factor U_i to the observation-process random effects. Default is c(intercept = -0.3, Z = -0.5).
sigma_q	Named numeric vector with elements intercept and Z giving the standard deviations of the residual observation-process random effects (square roots of the diagonal of Σ_q). Default is c(intercept = 0.3, Z = 0.2).
phi	Positive numeric scalar; decay rate of the OU process $\xi_i(t)$. The autocorrelation at lag Δt is $\exp(-\phi \Delta t)$. Larger values imply faster decay. Default is 0.1.
beta	Named numeric vector with elements intercept, Z, and X giving the outcome fixed-effect coefficients (excluding the time slope). Default is c(intercept = -2, Z = -0.5, X = 0.5).
beta_t	Numeric scalar; coefficient for the linear time effect in the outcome model. Default is 0.1.
theta	Named numeric vector with elements intercept and Z giving the outcome latent-link coefficients that couple the shared factor U_i to the outcome random effects. Default is c(intercept = 0.5, Z = 0.3).
Sigma_b	Numeric 2×2 positive-definite matrix; the residual covariance of the outcome random effects b_i (after removing the latent-factor contribution). Default is matrix(c(1, 0, 0, 2), 2, 2).
sigma_y	Positive numeric scalar; residual standard deviation of the outcome measurement error. Default is 1.

Value

An object of class "sim_GIVEHR_OU_PCL", which is a named list with the following components:

Visiting Process

The visit intensity for subject i is

$$\lambda_i(t) = \eta_i \exp(\gamma_0 + \gamma_Z Z_i + \gamma_X X_i),$$

where $\eta_i = \exp(\mu_0 + \sigma_\zeta U_i)$ is a log-normal frailty with $U_i \sim N(0, 1)$ and $\mu_0 = -\sigma_\zeta^2/2$. Visit times are generated as a homogeneous Poisson process with rate λ_i.

Observation Process

At each visit, the latent propensity for outcome observation is

$$\tilde{R}_{ij} = \alpha^\top X_{ij}^O + q_i^\top Z_{ij}^O + \xi_i(t_{ij}),$$

where $R_{ij} = I(\tilde{R}_{ij} > 0)$, $q_i \mid U_i \sim N(\delta U_i, \Sigma_q)$, and $\xi_i(t)$ is a stationary Ornstein–Uhlenbeck process with unit marginal variance and decay parameter ϕ.

Outcome Process

The outcome for subject i at time t is

$$Y_i(t) = \beta_0 + \beta_Z Z_i + \beta_X X_i + \beta_t t + b_i^\top Z_i^Y + \varepsilon_i(t),$$

where $b_i \mid U_i \sim N(\theta U_i, \Sigma_b)$ and $\varepsilon_i(t) \sim N(0, \sigma_y^2)$. The outcome is set to `NA` when $R_{ij} = 0$.

See Also

`GIVEHR`, `GIVEHR_GH`, and `GIVEHR_OU_PCL` for fitting models to the simulated data.

Examples

```
dat <- sim_data_gen(n = 30, seed = 42)
head(dat$long_data)
dat$params$beta
```

---

sim_ehr_data *Simulated Electronic Health Record (EHR) Longitudinal Data*

Description

A pre-generated longitudinal dataset from the joint model implemented in `sim_data_gen`, containing 2000 subjects with informative visiting and observation processes.

Usage

```
sim_ehr_data
```

Format

A `data.frame` with the following columns:

id Integer subject identifier.

Z Binary covariate (Bernoulli with probability 0.5).

X Continuous covariate (standard normal).

time Visit time; `NA` for subjects with no visits.

R Observation indicator: 1 if the outcome was recorded at the visit, 0 otherwise.

Y Longitudinal outcome; `NA` when `R = 0`.

C Censoring time (end of follow-up window, 60 for all subjects).

U True latent factor $U_i \sim N(0, 1)$ (included for evaluation; not available in real data).

eta True frailty η_i for the visiting process.

q_intercept, q_Z Observation-process random effects.

b_intercept, b_Z Outcome-process random effects.

Details

The data were generated with `sim_data_gen(n = 2000, seed = 2026)` and include a censoring-time column `C` set to the end of the observation window.

True parameter values used for the simulation are stored as an attribute and can be retrieved with `attr(sim_ehr_data, "true_params")`.

Source

Generated by `sim_data_gen` with default parameters and `n = 2000, seed = 2026`.

See Also

`sim_data_gen` for generating new datasets with custom parameters.

Examples

```
data(sim_ehr_data)
head(sim_ehr_data)

# Retrieve true parameters
true <- attr(sim_ehr_data, "true_params")
true$beta
```

---

summary.cimehr_bootstrap

Summarise a Bootstrap Result

Description

Displays bootstrap standard errors, percentile confidence intervals, and a summary of the replicate distribution for outcome coefficients.

Usage

```
## S3 method for class 'cimehr_bootstrap'
summary(object, digits = 4, ...)
```

Arguments

`object`	A `cimehr_bootstrap` object from `bootstrap`.
`digits`	Number of significant digits. Default is 4.
`...`	Ignored.

Value

`object`, invisibly.

See Also

`print.cimehr_bootstrap`, `bootstrap`

Examples

```
dat <- sim_ehr_data[sim_ehr_data$id <= 100, ]
fit <- IIRR_weighting(dat, visit_covs = c("Z", "X"))
bs  <- bootstrap(fit, data = dat, B = 5, seed = 1)
summary(bs)
```

`summary.EHRJoint` *Summarise an EHRJoint Fit*

Description

Displays estimates for the visiting, observation, and outcome processes.

Usage

```
## S3 method for class 'EHRJoint'
summary(object, level = 0.95, digits = 4, ...)
```

Arguments

`object`	An `EHRJoint` object.
`level`	Confidence level for Wald intervals. Default is 0.95.
`digits`	Number of significant digits. Default is 4.
`...`	Ignored.

Value

`object`, invisibly.

See Also

`print.EHRJoint`, `EHRJoint`

Examples

```
dat <- sim_ehr_data[sim_ehr_data$id <= 80, ]
fit <- EHRJoint(dat)
summary(fit)
```

summary.GIVEHR *Summarise a GIVEHR Fit*

Description

Displays estimates, sandwich standard errors, and Wald confidence intervals for all three stages: visiting process, observation process, and longitudinal outcome.

Usage

```
## S3 method for class 'GIVEHR'
summary(object, level = 0.95, digits = 4, ...)
```

Arguments

object	A GIVEHR object returned by GIVEHR.
level	Confidence level for Wald intervals. Default is 0.95.
digits	Number of significant digits. Default is 4.
...	Ignored.

Value

object, invisibly.

See Also

print.GIVEHR, GIVEHR

Examples

```
dat <- sim_ehr_data[sim_ehr_data$id <= 80, ]
fit <- GIVEHR(data = dat)
summary(fit)
```

summary.GIVEHR_GH *Summarise a GIVEHR_GH Fit*

Description

Displays estimates for all three stages of the Gauss–Hermite variant. Standard errors are shown where available.

Usage

```
## S3 method for class 'GIVEHR_GH'
summary(object, level = 0.95, digits = 4, ...)
```

Arguments

object	A GIVEHR_GH object returned by GIVEHR_GH.
level	Confidence level for Wald intervals. Default is 0.95.
digits	Number of significant digits. Default is 4.
...	Ignored.

Value

object, invisibly.

See Also

print.GIVEHR_GH, GIVEHR_GH

Examples

```
dat <- sim_ehr_data[sim_ehr_data$id <= 50, ]
fit <- GIVEHR_GH(data = dat, gh_points_U = 3, gh_points_q = 3)
summary(fit)
```

---

summary.GIVEHR_OU_PCL *Summarise a GIVEHR_OU_PCL Fit*

Description

Displays estimates for all three stages of the OU + PCL variant, including the OU decay parameter. Robust standard errors are shown where available.

Usage

```
## S3 method for class 'GIVEHR_OU_PCL'
summary(object, level = 0.95, digits = 4, ...)
```

Arguments

object	A GIVEHR_OU_PCL object returned by GIVEHR_OU_PCL.
level	Confidence level for Wald intervals. Default is 0.95.
digits	Number of significant digits. Default is 4.
...	Ignored.

Value

object, invisibly.

See Also

print.GIVEHR_OU_PCL, GIVEHR_OU_PCL

Examples

```
dat <- sim_ehr_data[sim_ehr_data$id <= 80, ]
fit <- GIVEHR_OU_PCL(data = dat, pair_type = "adjacent")
summary(fit)
```

`summary.IIRR_balancing` *Summarise an IIRR_balancing Fit*

Description

Displays visiting-process and outcome estimates under both balancing and MLE weights, including the sensitivity parameter.

Usage

```
## S3 method for class 'IIRR_balancing'
summary(object, level = 0.95, digits = 4, ...)
```

Arguments

`object`	An `IIRR_balancing` object.
`level`	Confidence level for Wald intervals. Default is 0.95.
`digits`	Number of significant digits. Default is 4.
`...`	Ignored.

Value

`object`, invisibly.

See Also

`print.IIRR_balancing`, `IIRR_balancing`

Examples

```
dat <- sim_ehr_data[sim_ehr_data$id <= 100, ]
fit <- IIRR_balancing(dat, visit_covs = c("Z", "X"))
summary(fit)
```

summary.IIRR_weighting *Summarise an IIRR_weighting Fit*

Description

Displays visiting-process and outcome estimates.

Usage

```
## S3 method for class 'IIRR_weighting'
summary(object, level = 0.95, digits = 4, ...)
```

Arguments

object	An IIRR_weighting object.
level	Confidence level for Wald intervals. Default is 0.95.
digits	Number of significant digits. Default is 4.
...	Ignored.

Value

object, invisibly.

See Also

print.IIRR_weighting, IIRR_weighting

Examples

```
dat <- sim_ehr_data[sim_ehr_data$id <= 100, ]
fit <- IIRR_weighting(dat, visit_covs = c("Z", "X"))
summary(fit)
```

summary.JMVL_Liang *Summarise a JMVL_Liang Fit*

Description

Displays visiting-process and outcome estimates.

Usage

```
## S3 method for class 'JMVL_Liang'
summary(object, level = 0.95, digits = 4, ...)
```

Arguments

object	A JMVL_Liang object.
level	Confidence level for Wald intervals. Default is 0.95.
digits	Number of significant digits. Default is 4.
...	Ignored.

Value

object, invisibly.

See Also

print.JMVL_Liang, JMVL_Liang

Examples

```
dat <- sim_ehr_data[sim_ehr_data$id <= 80, ]
fit <- JMVL_Liang(dat)
summary(fit)
```

---

summary.LI_IP_result *Summarise an LI_IP Fit*

Description

Displays imputation model details and outcome coefficients with standard errors when available.

Usage

```
## S3 method for class 'LI_IP_result'
summary(object, level = 0.95, digits = 4, ...)
```

Arguments

object	An LI_IP object.
level	Confidence level for Wald intervals. Default is 0.95.
digits	Number of significant digits. Default is 4.
...	Ignored.

Value

object, invisibly.

See Also

print.LI_IP_result, LI_IP

Examples

```
dat <- sim_ehr_data[sim_ehr_data$id <= 80,
                    c("id", "Z", "X", "time", "R", "Y", "C")]
fit <- LI_IP(data = dat, ip_method = "LMM",
             visit_covs = c("Z", "X"))
summary(fit)
```

---

summary.LMM_result *Summarise an LMM Result*

Description

Displays the fixed-effects coefficient table.

Usage

```
## S3 method for class 'LMM_result'
summary(object, level = 0.95, digits = 4, ...)
```

Arguments

object	An LMM_result object.
level	Confidence level for Wald intervals. Default is 0.95.
digits	Number of significant digits. Default is 4.
...	Ignored.

Value

object, invisibly.

See Also

print.LMM_result, LMM

Examples

```
dat <- sim_ehr_data[sim_ehr_data$id <= 100, ]
fit <- LMM(dat, fixed = Y ~ Z + X + time,
           random = ~ 1 | id, obs_indicator = "R")
summary(fit)
```

summary.MI_IP_result *Summarise an MI_IP Fit*

Description

Displays imputation settings and pooled outcome coefficients with standard errors when available.

Usage

```
## S3 method for class 'MI_IP_result'
summary(object, level = 0.95, digits = 4, ...)
```

Arguments

object	An MI_IP object.
level	Confidence level for Wald intervals. Default is 0.95.
digits	Number of significant digits. Default is 4.
...	Ignored.

Value

object, invisibly.

See Also

print.MI_IP_result, MI_IP

Examples

```
dat <- sim_ehr_data[sim_ehr_data$id <= 80,
                    c("id", "Z", "X", "time", "R", "Y", "C")]
fit <- MI_IP(data = dat, ip_method = "LMM",
             visit_covs = c("Z", "X"), m = 2, maxit = 5)
summary(fit)
```

summary.PairwiseLikelihood *Summarise a PairwiseLikelihood Fit*

Description

Displays model dimensions, convergence status, and the coefficient table with standard errors when available.

Usage

```
## S3 method for class 'PairwiseLikelihood'
summary(object, level = 0.95, digits = 4, ...)
```

Arguments

object	A PairwiseLikelihood object.
level	Confidence level for Wald intervals. Default is 0.95.
digits	Number of significant digits. Default is 4.
...	Ignored.

Value

object, invisibly.

See Also

print.PairwiseLikelihood, PairwiseLikelihood

Examples

```
dat <- sim_ehr_data[sim_ehr_data$id <= 100 &
                    sim_ehr_data$R == 1 & !is.na(sim_ehr_data$R), ]
fit <- PairwiseLikelihood(dat, formula = Y ~ Z + X + time,
                          pair_sample = 200)
summary(fit)
```

---

summary.SummaryStat_result

Summarise a SummaryStat Result

Description

Displays the regression coefficient table.

Usage

```
## S3 method for class 'SummaryStat_result'
summary(object, digits = 4, ...)
```

Arguments

object	A SummaryStat_result object.
digits	Number of significant digits. Default is 4.
...	Ignored.

Value

object, invisibly.

See Also

print.SummaryStat_result, SummaryStat

Examples

```
dat <- sim_ehr_data[sim_ehr_data$id <= 100, ]
fit <- SummaryStat(dat, formula = ~ Z + X)
summary(fit)
```

SummaryStat *Summary-Statistic Regression for Longitudinal Data*

Description

Reduces longitudinal data to a single summary statistic per subject (e.g., mean, median, min, max of the outcome), then fits a cross-sectional linear model on the summarized data. This provides a simple baseline analysis that ignores the within-subject correlation structure and visit process entirely.

Usage

```
SummaryStat(
  data,
  id = "id",
  outcome = "Y",
  stat = "mean",
  formula = ~1,
  cov_stat = "mean"
)
```

Arguments

data	A data.frame in long (panel) format with one row per subject-time combination.
id	Character string giving the name of the subject identifier column. Default is "id".
outcome	Character string giving the name of the outcome column. Default is "Y".
stat	Character string specifying the summary statistic to apply to the outcome within each subject. One of "min", "max", "mean", or "median". Default is "mean".
formula	A one-sided formula specifying the covariates for the linear model (e.g., ~ X + Z). The outcome (as summarized) is used as the response. Default is ~ 1 (intercept only).
cov_stat	Summary statistic to apply to time-varying covariates within each subject before fitting. Can be a single character string (applied to all covariates) or a named list mapping covariate names to statistic strings (e.g., list(X = "mean", W = "median")). Covariates not named in the list default to "mean". Default is "mean".

Details

The function proceeds as follows:

1. The outcome is aggregated to one value per subject using `stat` (e.g., subject-level means).
2. Each covariate in the formula is aggregated to one value per subject using `cov_stat`.
3. A standard linear model is fitted on the subject-level summary data.

This approach discards all within-subject temporal information and does not account for informative visiting or observation processes. It is included primarily as a naive benchmark for comparison with methods such as `GIVEHR`.

Value

A numeric matrix of regression coefficients, identical to the `coefficients` component of `summary(lm(...))`, with columns `Estimate`, `Std. Error`, `t value`, and `Pr(>|t|)`.

Examples

```
dat <- sim_ehr_data[sim_ehr_data$id <= 100, ]
fit <- SummaryStat(dat, formula = ~ Z + X)
print(fit)
```

---

vcov.GIVEHR *Variance–Covariance Matrix for GIVEHR Outcome Coefficients*

Description

Returns the sandwich variance–covariance matrix for the outcome model.

Usage

```
## S3 method for class 'GIVEHR'
vcov(object, ...)

## S3 method for class 'GIVEHR_OU_PCL'
vcov(object, ...)
```

Arguments

`object`	A `GIVEHR` or `GIVEHR_OU_PCL` object.
`...`	Ignored.

Value

Numeric matrix.

Examples

```
dat <- sim_ehr_data[sim_ehr_data$id <= 80, ]
fit <- GIVEHR(data = dat)
vcov(fit)
```

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